

Jalled Zain El Omar

Data Science Engineer | ML Engineering | MLOps

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SUMMARY

Final-year Data Science engineering student seeking a **PFE internship** in Data Science, ML Engineering, or MLOps. Hands-on experience building **LLM-powered agents, RAG systems, machine learning pipelines, and containerized services**. Skilled in Python, LangChain, LangGraph, FastAPI, Docker, MLflow, and data engineering, with additional professional experience in mobile software development and infrastructure monitoring. Interested in building reliable AI systems that connect **machine learning, software engineering, and production infrastructure**.

EDUCATION

- **Engineering Degree in Computer Science – Data Science** 2024 – Present
Ecole Supérieure Privée d'Ingénierie et de Technologies (ESPRIT)
- **Bachelor Degree in Information Technology** 2021 – 2024
Higher Institute of Technological Studies of Rades

TECHNICAL SKILLS

AI & Agents: Python, LangChain, LangGraph, Prompt Engineering, RAG, Autonomous Agents, pgvector
ML & DL: Scikit-learn, TensorFlow, PyTorch, MobileNetV2, Ensemble Methods, Stable Diffusion
MLOps & Infrastructure: MLflow, Docker, Docker Compose, Nginx, CI/CD, Zabbix, Elasticsearch, Kibana
Data & Databases: PostgreSQL, MongoDB, Oracle SQL, SQLite, Sentinel-2 / Google Earth Engine
Backend & Integration: FastAPI, REST APIs, Workflow Automation, MQTT, Webhooks, Event-driven Pipelines
Cloud & Systems: Linux, Docker Networking, JWT / Authentication, API Security
Development Tools: Git, GitHub, Flutter, ReactJS

CERTIFICATIONS

- **Applications of AI for Anomaly Detection** NVIDIA • Apr 2026
Autoencoders, anomaly scoring, network intrusion detection, threshold calibration
- **Fundamentals of Deep Learning** NVIDIA • Feb 2026
CNNs, transfer learning, GPU-accelerated model training

PROJECTS

- **InternHunter AI – Autonomous Internship Discovery Agent** / *Personal Project* Apr 2026
 - Architected a **LangGraph-based autonomous agent** orchestrating multi-step workflows for web search, semantic profile matching, ranking, and structured reporting with conditional branching and state management.
 - Integrated **Groq/Llama 3**, ChromaDB, and PostgreSQL for LLM reasoning, semantic memory, and persistent storage; connected Telegram and Notion to automate the end-to-end workflow.
 - Reduced manual internship search time from **2+ hours/day to near-zero** by automating discovery across 5+ platforms targeting the French and European market.
- **AgriSmart – AI-Powered Precision Agriculture Platform** / *Solution Architect, Team of 6* Jan – May 2026
 - Architected a **microservices platform** with 6 FastAPI services, Docker Compose, and Nginx; built a **RAG irrigation assistant** using LangChain, Llama 3, and pgvector for grounded farmer queries.
 - Deployed on-device plant disease detection using **MobileNetV2 TFLite Int8** on 87K+ images with sub-3s offline inference; integrated **Sentinel-2 / Google Earth Engine** for geospatial crop analytics and led onboarding for 5 team members.
- **End-to-End MLOps Pipeline – Drug Classification** / 📁 *GitHub* Nov – Dec 2025
 - Trained a **Random Forest classifier** with SMOTE on an imbalanced medical dataset, achieving a **0.89 F1-score**; served predictions via FastAPI with sub-100ms latency and tracked experiments with **MLflow**.
 - Containerized the ML stack with Docker Compose and integrated **Elasticsearch + Kibana** for application observability and monitoring.

PROFESSIONAL EXPERIENCE

- **Infrastructure Monitoring Intern** On-site
Delice Danone / Tunisia Jul – Sept 2026
 - Built an end-to-end **Zabbix 7.0 / Hyper-V monitoring pipeline** using Docker and PostgreSQL, with VM auto-discovery (LLD), custom **PowerShell UserParameter** scripts, PSK encryption, SLA triggers, and alerting dashboards.
 - Developed a **Python metrics collector** pulling live Zabbix API telemetry into SQLite, creating a data pipeline for future **anomaly detection** on infrastructure monitoring data.
- **Mobile Developer – Part-time** Remote
Kinko Booking / Tunisia Dec 2025 – Present
 - Developed and maintained production mobile features in **Flutter**, applying clean architecture and modular application design.
 - Implemented application functionality including **Firebase Cloud Messaging, GoRouter navigation, API integration**, and state/-data flow across the mobile application.
 - Contributed to the mobile development workflow through **Git-based collaboration and CI/CD**, supporting application testing and release processes.
- **Data Engineering Intern** On-site
Proservices Training Company / Ariana, Tunisia Jun – Aug 2025
 - Designed and implemented a **PostgreSQL analytics backend** capturing learning engagement metrics for **100+ active users**, with optimized indexing for query performance.
 - Built scalable **RESTful services using NestJS** and applied MVVM architecture to maintain structured data flows between backend APIs and the Flutter mobile client.